CHAPTER 8

Our planet Earth



Outcome

Earth and space sciences

4.1

Relate the occurrence of natural events to atmospheric changes and movements in the Earth's crust and mantle

Stimulus questions

1. Describe the shape of the Earth.
2. What is the difference between rocks and soil?
3. What is the difference between the air we breathe in and the air we breathe out?
4. How high do we need to travel before we reach 'space'?
5. Why do winds happen?

Unit 8.1 What on Earth?

For the last 4.5 to 6 billion years, this lump of rock we call Earth has been revolving around the sun.

It began as a ball of molten matter, eventually cooling at the surface until it hardened and became solid. Since then the surface has been shaped by wind, rain, ice and heat. The planet has also produced an atmosphere that has the temperature and the gases needed for all living things to live, grow and prosper.

With a diameter of only 12 756 km, the planet Earth is a small one.

This is about twelve times the distance from Melbourne to Sydney and is less than the distance from Melbourne to Los Angeles (13 200 km), a distance aircraft fly non-stop each day.

Although it appears from space as a perfect sphere, Earth is actually a little squashed at the poles and slightly fatter at the equator because of its spin.

29% of the surface is land while the other 71% is water (97% of the water is salt water with another 2% stored as ice in Antarctica and Greenland).

Journey to the centre of the Earth

If we dug a hole through to the centre of the Earth, we would find that the Earth is made up of **layers**.

The crust

This layer (also called the **lithosphere**) contains the mountains and the seas and is the rock we dig up in mines. Covered by a thin layer of soil and sand, the crust is a solid 'shell' made mostly of solid rock. It is thickest under the continents (about 33 km thick) and thinnest under the sea (11 km thick). It is extremely thin when compared to the total volume of the Earth and is like a postage stamp stuck on a basketball.

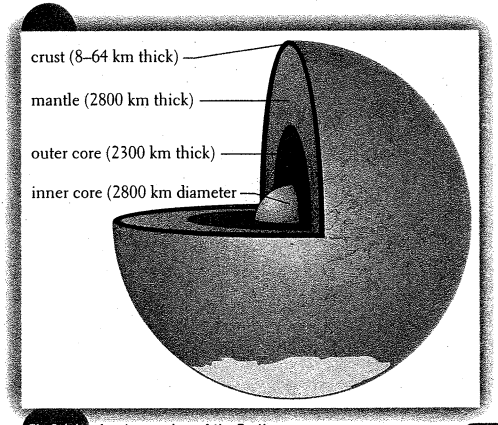


Fig 8.1.1 A cutaway view of the Earth

WS 8.1

The temperature increases from an average of 20°C at its surface, to about 500°C at its maximum depth.

The mantle

This layer is 2800 km thick with temperatures of 500°C near the crust and 3000°C nearer the core.

Digging here would be impossible since the rock is mostly molten, moving continually and affecting the crust above. This is the source of much **volcanic activity** and **earthquakes**.

The outer core

This is 2300 km of continually moving molten iron and nickel. These metals give the Earth its north and south **poles** and its **magnetic field**. This field acts as a '**cosmic shield**', protecting us by deflecting large doses of **cosmic rays** from the sun.

The temperature of the outer core varies from 4000°C to 6000°C.

The inner core

The remaining 1400 km is through the inner core where temperatures range from 4000°C to 7000°C

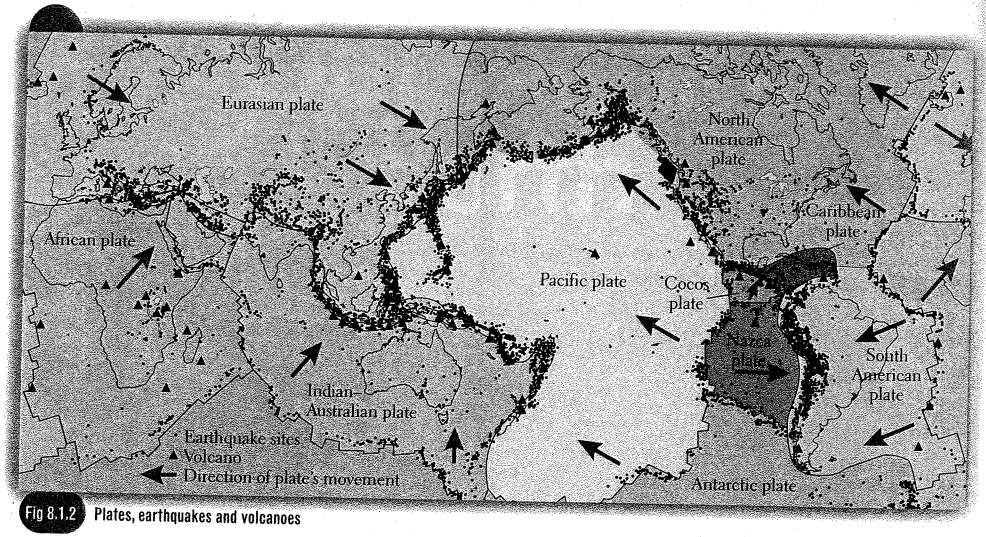


Fig 8.1.2 Plates, earthquakes and volcanoes

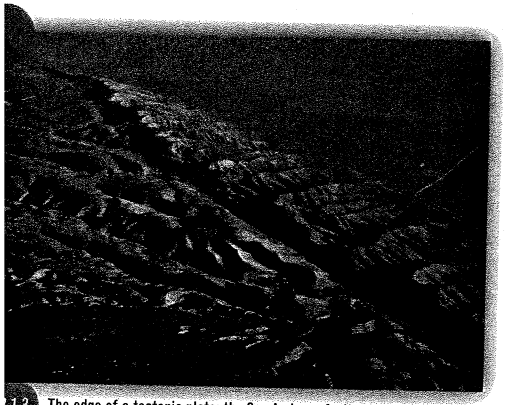
At these temperatures the iron and nickel that make up the core should be molten but the extreme high pressures from the layers above keep these metals solid.

Crashing plates

The crust of the Earth looks as if it could never move, but bits of it are moving all the time. The crust is broken into huge slabs of rock called **plates**. Below this is the continually moving molten rock of the mantle. The plates 'float' on the **convection currents** of the mantle and eventually crash into one another, break away from each other or slip along or under each other.

This sliding is not smooth: rocks stick and jam. Pressure builds until the rocks can take no more. There is a sudden explosive slippage and an earthquake happens, its damage being the most severe at the edges of the plate.

Shifting plates also suggest that the continents must be slowly moving too. This idea was first suggested in 1912 but was hardly believed at first since it sounded so ridiculous. Clues that the continents were once stuck together as super-continents that have slowly shifted to their current positions, come from the shapes of the continents themselves and the types of rocks and plant life found on them.



.1.3 The edge of a tectonic plate: the San Andreas fault causes up to five tremors a day through California

Science snippet

A changing Earth

A map of the world of the distant future would look very different to that of today. Because the continents are shifting, Australia is drifting north at about 5 cm per year! The Mediterranean Sea is slowly being squeezed shut, the Atlantic Ocean is getting wider and the Himalayan mountains are getting higher!

The idea of shifting plates (and not just continents) was first introduced in 1969 and is called the **theory of plate tectonics**.

Prac 1 p. 208

Unit 8.1 Questions

1 Copy the following, correcting any incorrect statements so they become true:
- a The inner core of the Earth is solid.
- b The iron and nickel in the crust give the Earth its magnetic field.
- c Most of the Earth's surface is made of land.
- d Mines are often deep enough to go into the mantle.
- e The Earth is spherical like a ball but is slightly flattened at the poles.

2 On a diagram of planet Earth show:
- a the poles
- b the equator
- c the tropics of Cancer and Capricorn
- d the direction it spins

3 What percentage of the Earth's surface is:
- a water?
- b land?

4 Where is most of the Earth's:
- a salt water?
- b fresh water?

5 1% of fresh water is missing in the numbers given on page 205. Where on Earth could this water be?

6 Which of the layers of the Earth is:
- a the thickest?
- b the hottest?
- c mainly made of iron and nickel?
- d liquid?
- e solid?

7 How would life on Earth be affected if there was no magnetic field?

8 Where is the crust the:
- a thickest?
- b thinnest?

9 Scientists want to explore the Earth's structure by digging a hole deep through the crust into the mantle. What are the advantages and disadvantages of choosing Central Australia for the dig?

10 A digging machine has been invented that can dig 10 km per day. How long would it take to cut through each layer of Earth and how long would it take to reach the other side?

11 Explain how the distance from Melbourne to Los Angeles could be further than the diameter of Earth.

12 Australia is drifting northwards at 5 cm per year. This means that Melbourne will eventually be where Sydney is now! If the distance between the two cities is 900 km, calculate how long this will take.

13 Use this information to plot a graph showing the temperature for every kilometre as we dig into the crust.

Depth (km)	0	2	4	6	8
Temperature (°C)	20	87	153	220	286

14 Use the graph from question 13 to estimate the temperature at these depths:
- a 1 km
- b 5 km
- c 10 km
- d 20 km

15 At what depth would the temperature be roughly:
- a 50°C
- b 100°C
- c 200°C
- d 300°C

Unit 8.1 Research/ Extension

1 Make a scale model of Earth showing its layers. Mark the poles, equator, the hemispheres, the tropics of Capricorn and Cancer and the direction of the Earth's spin.
2 Imagine digging a hole from where you live straight through Earth to the other side. Use a globe to estimate where the tunnel would emerge.
3 What are the seven continents of the world and what is the highest mountain on each?
4 Use an atlas to find the spot on Earth where you would expect the crust to be:
   a the thickest?
   b the thinnest?
   c the thinnest while you are standing on land?
5 Find the depth of the deepest holes ever drilled on Earth.
6 Find maps of the ancient supercontinents of Pangaea and Gondwanaland.

Unit 8.1 Prac 1
The crust is like an eggshell

You will need
Fresh hard-boiled egg

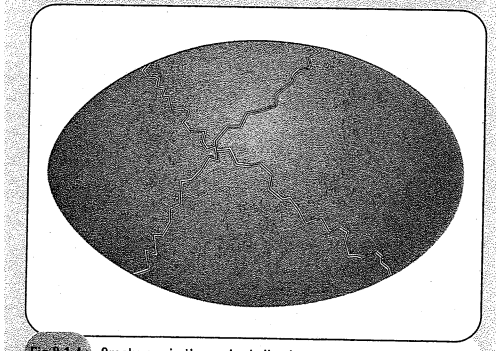


Fig 8.1.4 Cracks are in the egg's shell only

What to do
1 Tap the egg firmly so that the shell cracks, but do not peel off the shell.
2 Squeeze the egg gently but not enough to destroy the egg.
3 Try to slide one piece over another.

Questions
1 What happens to the cracks in the shell?
2 Record your observations.
3 How is this cracked egg similar to the Earth?
4 What would an ant standing on one of the cracks experience during the experiment?

Unit 8.2 Earthquakes



Fig 8.2.1 San Francisco, 1989

Although more than one million earthquakes hit Earth every year, most are so small that almost no-one notices. The scientists who study them (called **seismologists**) know even when small earthquakes happen by measuring the movement of the Earth's crust with an instrument called a **seismometer**. The graph it makes (a **seismograph**) gives information about how serious the quake was. It can also be used to determine where the quake began (the **focus**) and what point on Earth was directly above it (called the **epicentre**). The focus can be as deep as 100–700 km below the surface. WS 8.2

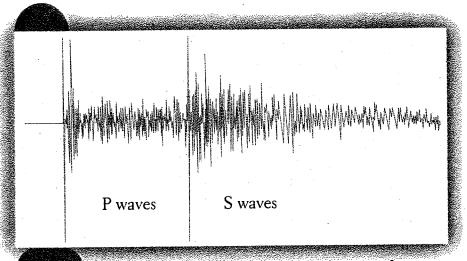


Fig 8.2.2 A seismograph is a drawing of the shock waves of an earthquake. P waves arrive first. They are 'longitudinal' waves and push and pull the Earth. S waves arrive next. They are 'transverse' waves and shake the surface.

Richter and Mercalli

Scientists use seismographs to estimate the **energy** of an earthquake at its epicentre.

An earthquake's strength is measured on a scale from 0 to 9 devised by American Charles **Richter** in 1935. Each number is ten times worse than the number before. This means that an earthquake of strength 8 has ten times the energy of an earthquake of strength 7, which then has ten times more energy than a 6. Any quake above 6 on the Richter scale must be considered dangerous since major damage can be expected.

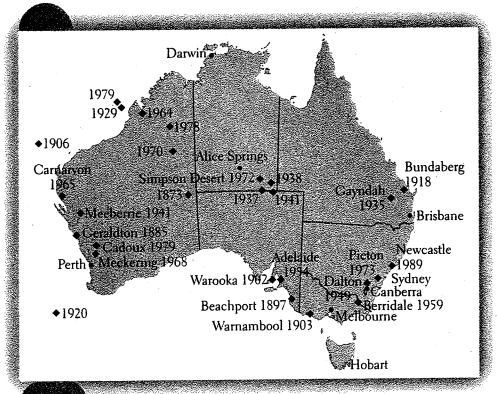


Fig 8.2.3 Earthquakes stronger than 6 on the Richter scale

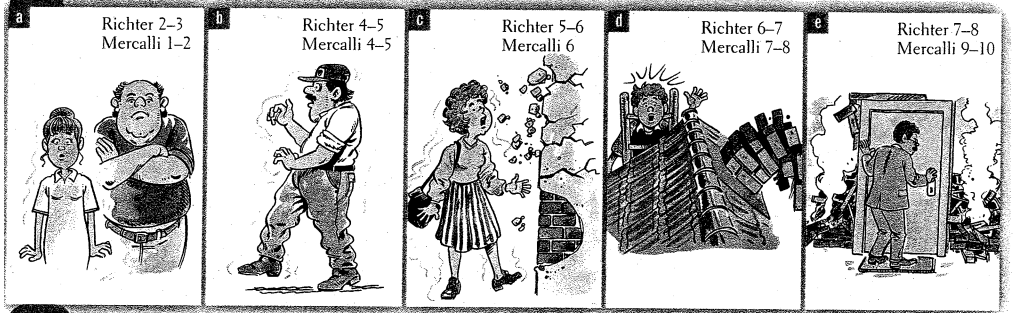


Fig 8.2.4 a Very feeble—noticed only by sensitive people; b Moderate—felt by people who are moving about; c Strong—slight damage to buildings; d Destructive—chimneys fall; e Disastrous—many buildings are destroyed

Luckily only nineteen such quakes over 6 on the Richter scale have been recorded in Australia.

If no instruments are present to measure the quake, the **Mercalli** scale is used instead. WS 8.3

Damage caused	Richter	Mercalli	Average number per year
Felt by seismographs only	1–2	0	More than 500 000
Felt by very few people	2–3	1–2	100 000 to 500 000
Felt by people in tall buildings; hanging objects swing	3–4	2–3	10 000 to 100 000
Felt and heard by most; parked cars rock, crockery rattles, walls crack	4–5	4–5	1000 to 10 000
Felt by all; some panic, furniture moves, difficult to walk	5–6	6	200 to 1000
Difficult to stand; chimneys and some buildings collapse; cracks in the ground	6–7	7–8	20 to 200
General panic; deep cracks in the ground, most buildings collapse, rail lines twist, dams break	7–8	9–10	10 to 20
Total destruction; valleys fill with mud from landslides and flood	8–9	11–12	0 to 10

Other earthquake nasties

Large earthquakes move large slabs of crust around. These slabs take some time to settle and cause smaller earthquakes called **aftershocks** when they do. Although generally smaller than the first quake, aftershocks can be extremely dangerous, particularly if buildings were made unstable by the first quake.

Gas pipes can easily be broken in an earthquake. One spark or flame can turn the wreckage into a furnace. In 1906 San Francisco was hit by a quake estimated at 8.3 on the Richter scale. 60% of the damaged city (520 city blocks!) then burnt to the ground in the following days.

An earthquake with its **epicentre** under the ocean floor can cause a wave to be formed. Although it may be only 2 m high, it travels at incredible speed and grows as it enters shallow water near the shore. This wave is called a **tsunami** (pronounced soon-army) and can grow up to 30 m tall.

WS 8.4 WS 8.5

Science snippet
The worst ever

Earthquakes were first recorded in China around 1000 BC. The most devastating happened in 1556 at Hsian where most of the people had dug cave-like homes into the hills. These collapsed, burying them alive. Landslides, floods, famine and disease claimed more. A total of 830 000 are thought to have died.

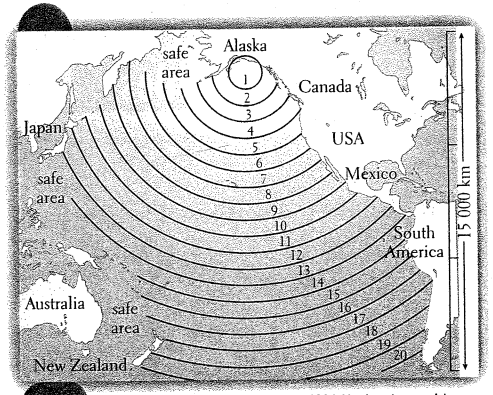


Fig 8.2.5 The number of hours taken for the 1964 Alaskan tsunami to cross the Pacific Ocean

In 1998 an underwater quake of 7 on the Richter scale caused a 15 m high wave to hit the north coast of Papua New Guinea killing an estimated 3000 people. On an even larger scale, a tsumani which hit Japan in 1892 caused the death of 27 000 people.

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Science snippet
Animals predicting earthquakes?

In China in 1975, animals began to act strangely. Snakes left their burrows as if in fear and dogs barked wildly at nothing. Authorities thought the animals could sense an earthquake approaching. They warned that a major quake would occur in the next 6 months. Townsfolk were evacuated to the country. Soon after, a quake did occur and because of the precautions few died.

Another quake struck 2 years later, killing 240 000 people. Were the animals lying down on the job this time?

Unit 8.2 Questions

1 What do you call a scientist who studies earthquakes?
2 What's the difference between the epicentre and the focus of an earthquake?
3 Copy the following, correcting any incorrect statements so they become true:
   a Aftershocks are huge waves caused by earthquakes.
   b A quake of strength 5 on the Richter scale is double the strength of a 4.
   c Earthquakes are caused by plates slipping.
   d The focus of a quake is the exact point where an earthquake starts.
   e Tsunamis are huge when in deep water.
4 What are two reasons that most quakes are not felt?
5 Where do quakes occur?
6 Explain why earthquakes occur.
7 What are aftershocks and what causes them?
8 Why are aftershocks often more dangerous than the first quake?
9 What value on the Richter scale would you call a 'serious' quake? Explain why you have chosen this number.
10 A tremor is an earthquake that can be felt but does little damage. What would it be on the Richter and Mercalli scales?
11 Superquakes are ones that are more than 8 on the Richter scale.
   a What damage can be expected when superquakes hit?
   b How often do they occur per year?
12 Make a series of sketches to explain the damage that can be expected for each level of the Mercalli and the Richter scales.
13 There are almost no photographs or videos anywhere of tsunamis. Suggest why.
14 In 1989 a major quake of strength 7.1 hit San Francisco, killing 62. Fires were not a great concern. Make three suggestions why this quake was not as deadly or as damaging as the one in 1906.

10.4

Windows into the Earth

When you look out your bedroom window you see only a tiny part of the world. An Inuit child looking out a frosty window in northern Canada would see a very different view. Each window gives only a small amount of information, but as more and more 'windows' are added together, the view of the world becomes more complete.

Talk about

Why do we need the 'views from many windows' to get a complete view of the world?

High temperatures and pressures have so far prevented scientists from drilling more than about 10 kilometres into the Earth. Because of this, we have to rely on the Earth's 'natural windows' to provide us with information about the planet's interior. These windows are as yet very small, but through them we are steadily increasing our knowledge. Although nobody has seen the inside of the Earth, we have a good idea what it looks like. What are some of these 'windows'?

Earthquakes provide one window into the Earth. Scientists called **seismologists** who study earthquakes tell us that the inside of the Earth has several layers, some of which are more solid than others. Volcanoes are another kind of 'window'. **Vulcanologists** are able to study material from the upper part of the mantle when it comes to the surface during a volcanic eruption. **Meteorites** are thought to be lumps of 'leftover building material' from the time of the formation of the planets. Some scientists believe that these rocks consist of similar chemicals to those inside the Earth. We also know that the Earth has a magnetic field around it, and this offers another clue. Many scientists suggest that the magnetic field is caused by a layer of liquid iron inside the Earth—most probably the outer core.

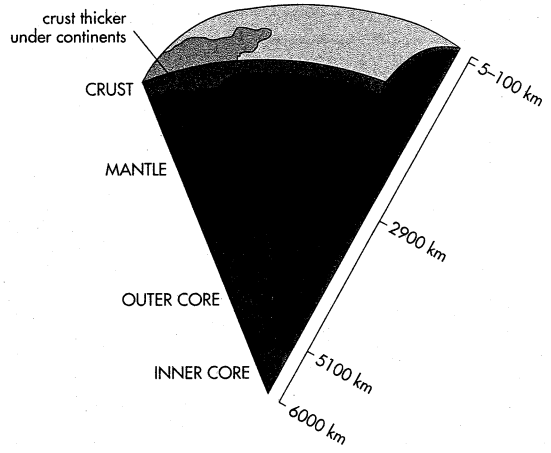


FIGURE 10.9
The layers inside the Earth.

USING SCIENCE THE LAYERS OF THE EARTH

Work in five groups to make a rough scale model of the interior of the Earth. You will need to work in a large outdoor area such as on the school oval.

WHAT YOU NEED

- 5 large pieces of stiff cardboard (poster size)
- 5 position markers (witches' hats or buckets)
- 5 thick marking pens

WHAT TO DO

1 Each group chooses one of the layers and makes a very large, highly visible sign for it. For example:
TOP OF CRUST
TOP OF MANTLE
TOP OF OUTER CORE
TOP OF INNER CORE
CENTRE OF EARTH

2 Take the signs, position markers, and one book per group outside to a large open area.

3 Make a starting point at one end of the area. The 'TOP OF CRUST' group stays here, holding up their sign and placing the marker on the ground at their feet.

4 The 'TOP OF MANTLE' group stands in front of the first group, as close as possible. They hold up their sign and put their marker at their feet.

5 The other groups take 30 large paces from 'TOP OF MANTLE'. The 'TOP OF OUTER CORE' group stays here with their sign and marker.

6 The remaining two groups take 20 large paces and 'TOP OF INNER CORE' stays here.

7 The last group goes to the 'CENTRE OF THE EARTH', a further 15 paces away.

8 Everyone should hold up their signs and look at the model they have made. Leave the signs and markers in place, and go back to the beginning.

POINTS TO CONSIDER

1 The people at the top of the crust would be most affected by earthquakes. Notice how thin the crust is: humans have not even drilled through this layer!

2 The group at the top of the mantle were standing where volcanic eruptions start.

3 The people of the outer core were probably standing in the liquid iron and nickel that are thought to generate the Earth's magnetic field.

4 The group at the inner core were so squashed by the weight of everything else above that they were rock solid.

5 At the centre of the Earth, nearly 6000 kilometres down, the last group was sizzling at over 4000°C.

EXTENSION

Find out more about the different layers of the Earth and produce your own large, colourful wall chart. For each layer, include information such as temperatures and what the layer is made of. You might even like to make part of the wall chart three-dimensional, moveable, or perhaps 'pop-up'.

Questions

1 What does a vulcanologist study?

2a List four natural occurrences that can give scientists clues to the interior of the Earth.

b Explain how each of these clues adds to the 'big picture'.

3a Which is the thickest layer of the Earth?

b Which is the thinnest?

Think about

A famous man, Will Durant, once said:

'Civilisation exists by geological consent, subject to change without notice.'

Discuss this statement with a friend, your group, or the whole class. What do you think he meant? What 'changes without notice' could be brought about by activity within the crust and mantle?

10.5

Earthquakes and tsunamis

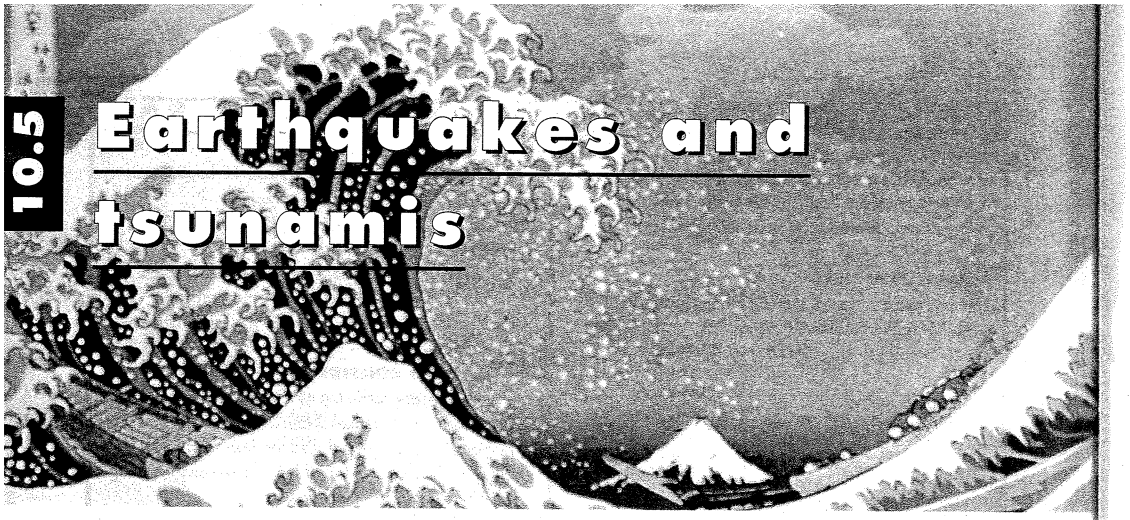


FIGURE 10.10
In 1946 a young Hawaiian girl stood in disbelief at the sight from her clifftop home above Hilo Bay. As if drawn by some giant vacuum cleaner, the sea water below her was sucked away leaving countless stranded fish flapping about on the exposed sea floor. As if in slow motion, a huge wall of grey water 30 metres high surged back to the shore. It swamped the town below her, killing 96 people and leaving the whole area in ruins.

Talk about
Imagine that a 30-metre wave was 30 seconds away from your home. What would you do?

The event described in the caption was a **tsunami** (Japanese for 'harbour wave'). An earthquake under the ocean is the most common cause of tsunamis, but the biggest and most destructive tsunamis yet recorded have been started by underwater volcanic eruptions.

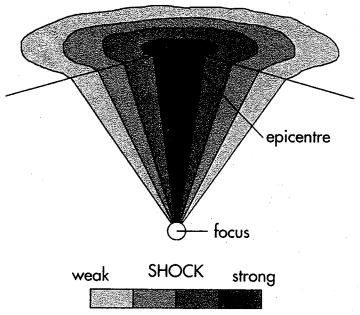


FIGURE 10.11
When an earthquake occurs, shock waves travel in all directions from the focus, reaching the epicentre first.

Movements of the Earth's crust are called **earthquakes**. Earthquakes start in the solid crust and upper mantle of the Earth, at depths between 70 kilometres and 700 kilometres. Parts of the Earth's crust are under enormous pressure. When this pressure becomes too much, the rocks may move or break, sending out shock waves called **seismic waves** in every direction. This results in ground movement which can cause great damage.

The point at which an earthquake starts is called the **focus**. The point on the Earth's surface directly above the focus is called the **epicentre**. The strongest shaking is usually felt at the epicentre.

Measuring the strength of earthquakes

About 150 000 earthquakes are recorded each year. **Seismographs**, machines that measure the amount of ground shaking, produce **seismograms**, graphical records of the size and duration of the earthquake. Computers can be used to find the exact location and depth of the focus.

The strength of an earthquake is often described in terms of numbers on the Richter scale, as shown in the table on the following page.

TABLE 10.1 The Richter scale.

Richter scale measurement	Earthquake effects
3	Felt slightly in areas near epicentre. No damage.
4	Felt by most people up to several kilometres from earthquake. Some objects upset.
5	Strongly felt. Some damage to weak buildings.
6	Moderately destructive. Some severe damage to weak buildings.
7	Major earthquake. Destruction of many buildings.
8	Very destructive. Total destruction of structures.

Each increase of 1 on the Richter scale corresponds to a ten-fold increase in energy. For example, an earthquake measuring 6 on the Richter scale is 10 times more powerful than an earthquake measuring 5.

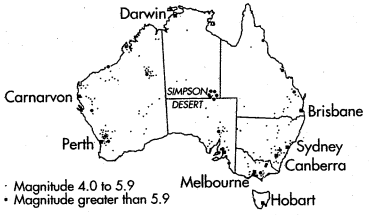


FIGURE 10.12
Known earthquake sites in Australia.

Australia has very few large earthquakes. Those earthquakes that do occur usually don't cause much damage because our country is so lightly populated. The Newcastle earthquake of 1989 was an exception. Twelve people lost their lives, and damage to buildings in this heavily populated area was extensive. Yet the earthquake itself was only moderate, measuring 5.6 on the Richter scale.

USING SCIENCE MAKE A MODEL SEISMOGRAPH

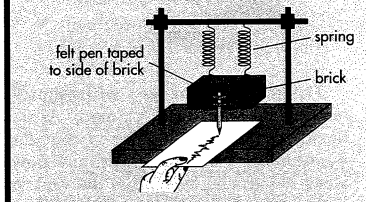


FIGURE 10.13
A model seismograph.

WHAT TO DO

1 Use the diagram opposite as a guide to designing and building your own seismograph. Try out different materials until it works.

2 Test your seismograph by bumping the table or jumping on the floor while another student pulls the paper past the pen. See what other tests you can come up with.

3 Develop your own scale for measuring the strength of your bumps and thumps.

Questions

1a What causes tsunamis?

b Tsunamis are often called 'tidal waves'. Is this an accurate description? Explain your answer.

2 Use the map in Figure 10.12 to answer the following questions.

a Where have the biggest earthquakes in Australia occurred?

b Would these earthquakes have caused a lot of damage? Why?

c Are there any capital cities that could claim to be 'earthquake-free'? Which one(s)?

3 If the tsunami that hit Hawaii in 1946 had travelled 3200 kilometres in 5 hours, how fast did it travel (in km/h)?

4 Use the Richter scale (Table 10.1) to suggest what damage you would have expected in the Newcastle earthquake.

FACT FILE

The greatest death toll due to an earthquake occurred in China in 1556 when approximately 830 000 people were killed.

Unit 8.2 Research/ Extension

1 Construct a time line of 'serious earthquakes of the last century'.
2 For the Alaskan tsunami of 1964, find:
   a the maximum distance it travelled
   b what cities would have been affected
   c the speed it travelled cross the Pacific
3 Find out about S, P and L waves, what damage they do and how they travel.

CREATIVE WRITING

Eyewitness news

Choose one earthquake from the last hundred years. You were a resident of that city. Write an account of your experiences on the day of the quake and the next week.

You must include:

- 'eyewitness' accounts of the quake
- how you protected yourself
- its strength on the Richter and Mercalli scales
- the immediate destruction
- the problems that the survivors encountered in the following week
- the final toll in life and property damage

You can write your account as:

- a series of diary entries
- a Powerpoint presentation of the week
- a series of newspaper front pages
- a Word document with hyperlinks

Unit 8.2 Practical activities



Unit 8.2 Prac 1
Building a seismometer

You will need

Retort stands, bossheads and clamps, spring, heavy weight, adhesive tape, felt pen

What to do

1 Construct the contraption in the figure.
2 Slowly pull the paper while creating 'earthquakes' by thumping the bench on top and at the end.

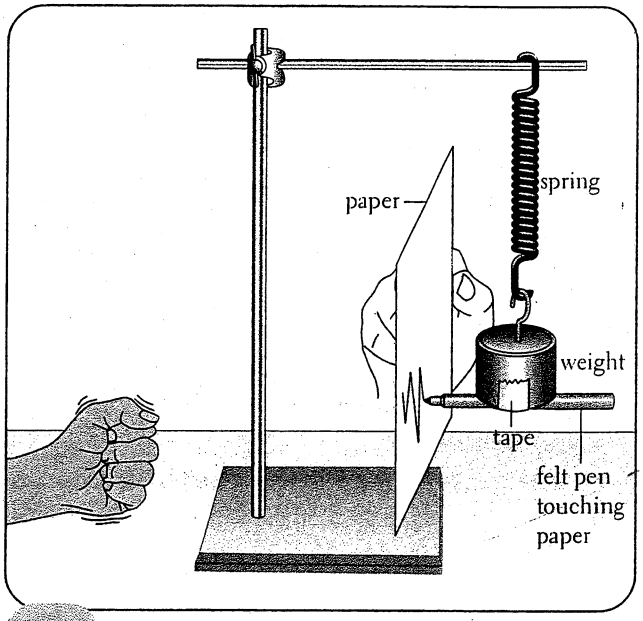


Fig 8.2.6 Modelling a seismograph

Questions

1 Which movements caused the seismographs to work?
2 Why didn't the graph stop after the 'quake' did?